

## Curriculum Version 1.0

### B.Tech. (Computer Science and Engineering-AIML)

(For AY2024-25 onwards)

| Credit distribution among baskets<br>(Minimum credits to be earned = 160) |                               |                     |
|---------------------------------------------------------------------------|-------------------------------|---------------------|
| Sl. No.                                                                   | Baskets                       | Credit Contribution |
| 1                                                                         | School Core                   | 58                  |
| 2                                                                         | Program Core                  | 41                  |
| 3                                                                         | Discipline Elective           | 43                  |
|                                                                           | Common Electives Basket       | 10                  |
|                                                                           | Specialized Elective Baskets* | 33                  |
| 4                                                                         | Open Elective                 | 18                  |
| Total Credits                                                             |                               | 160                 |

| Specialized Elective Baskets* |                                              |                            |
|-------------------------------|----------------------------------------------|----------------------------|
| Sl.No.                        | Name of the Basket                           | Minimum Credit Requirement |
| 1                             | Artificial Intelligence and Machine Learning | 12                         |
| 2                             | Cyber Security                               | 3                          |
| 3                             | Data Analytics                               | 3                          |
| 4                             | IoT                                          | 3                          |
| 5                             | Cloud Computing                              | 3                          |
| 6                             | Robotics                                     | 3                          |
| 7                             | Big Data                                     | 0                          |
| 8                             | Fullstack Development                        | 3                          |
| 9                             | DevOps                                       | 3                          |
| Total*                        |                                              | 33                         |

*\* Remaining Credits, if any, to satisfy Specialized Elective Baskets credit requirement, can be taken from any Specialized Elective Basket*

| Sl. No.                                                                                                                           | Course Code | Course Title                                           | T | P | Credits   |
|-----------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------------------------------------------|---|---|-----------|
| <b>A. School Core - Minimum Credits to be earned from this basket (A+B+C+D+E+F+G) =</b>                                           |             |                                                        |   |   | <b>58</b> |
| 1                                                                                                                                 | CSE1018     | Object Oriented Programming with Java                  | 2 | 4 | 4         |
| 2                                                                                                                                 | CSE1017     | Programming in C and C++                               | 2 | 4 | 4         |
| 3                                                                                                                                 | CSE2001     | Data Structures and Algorithms                         | 3 | 2 | 4         |
| 4                                                                                                                                 | MAT1001     | Linear Algebra and Calculus                            | 3 | 0 | 3         |
| 5                                                                                                                                 | MAT1002     | Differential Equations and Transform Techniques        | 3 | 0 | 3         |
| 6                                                                                                                                 | MAT1013     | Statistics and Probability                             | 2 | 2 | 3         |
| 7                                                                                                                                 | MAT1012     | Numerical Techniques                                   | 2 | 2 | 3         |
| 8                                                                                                                                 | MGT1101     | Digital Entrepreneurship                               | 2 | 0 | 2         |
| 9                                                                                                                                 | BIT1002     | Basic Human Nutrition                                  | 2 | 0 | 2         |
| 10                                                                                                                                | ECE1008     | Innovation Project Using Arduino                       | 0 | 2 | 1         |
| 11                                                                                                                                | ECE1009     | Innovation Project Using Raspberry Pi                  | 0 | 2 | 1         |
| 12                                                                                                                                | CSE4001     | B.Tech. Capstone Project                               | - | - | 4         |
| 13                                                                                                                                | CSE4002     | B.Tech. Internship                                     | - | - | 6         |
| <b>B. Electrical and Electronics Basket - Min. credits to be earned from this basket =</b>                                        |             |                                                        |   |   | <b>4</b>  |
| 1                                                                                                                                 | ECE1002     | Elements of Electronics Engineering                    | 3 | 2 | 4         |
| 2                                                                                                                                 | ECE1001     | Fundamentals of Electrical and Electronics Engineering | 3 | 2 | 4         |
| <b>C. Modern Physics Basket - Min. credits to be earned from this basket =</b>                                                    |             |                                                        |   |   | <b>3</b>  |
| 1                                                                                                                                 | PHY1001     | Physics of Opto-electronic Devices                     | 2 | 2 | 3         |
| 2                                                                                                                                 | PHY1002     | Semiconductor Physics                                  | 2 | 2 | 3         |
| <b>D. English Basket - Min. credits to be earned from this basket =</b>                                                           |             |                                                        |   |   | <b>2</b>  |
| 1                                                                                                                                 | ENG1002     | Communicative English                                  | 0 | 2 | 1         |
| 2                                                                                                                                 | ENG1003     | Professional English                                   | 0 | 2 | 1         |
| <b>E. Foreign Languages Basket - Min. credits to be earned from this basket =</b>                                                 |             |                                                        |   |   | <b>1</b>  |
| 1                                                                                                                                 | FRE1001     | Basic French                                           | 2 | 0 | 2         |
| 2                                                                                                                                 | GER1001     | Basic German                                           | 2 | 0 | 2         |
| 3                                                                                                                                 | FRE1002     | Communicative French                                   | 0 | 2 | 1         |
| 4                                                                                                                                 | GER1002     | Communicative German                                   | 0 | 2 | 1         |
| <b>F. Behavioural Science Basket (All Courses in this basket are mandatory) - Minimum Credits to be earned from this basket =</b> |             |                                                        |   |   | <b>2</b>  |
| 1                                                                                                                                 | PSY1001     | Understanding Self for Effectiveness                   | 0 | 2 | 1         |
| 2                                                                                                                                 | PSY1002     | Dynamics of Human Behaviour                            | 0 | 2 | 1         |
| <b>G. Soft Skills Basket (All Courses in this basket are mandatory) - Minimum Credits to be earned from this basket =</b>         |             |                                                        |   |   | <b>3</b>  |
| 1                                                                                                                                 | SSK2002     | Being Corporate Ready                                  | 0 | 2 | 1         |
| 2                                                                                                                                 | SSK3001     | Problem Solving through Aptitude                       | 0 | 2 | 1         |
| 3                                                                                                                                 | CSE3050     | Programming Skills for Employment                      | 0 | 2 | 1         |
| <b>H. Python Basket - Min. credits to be earned from this basket =</b>                                                            |             |                                                        |   |   | <b>2</b>  |
| 1                                                                                                                                 | CSE1005     | Programming in Python                                  | 0 | 4 | 2         |
| 2                                                                                                                                 | CSE1037     | Programming in Python                                  | 2 | 2 | 3         |
| <b>Kannada Basket</b>                                                                                                             |             |                                                        |   |   | <b>1</b>  |
| 1                                                                                                                                 | KAN1005     | Kannada Kali                                           | 0 | 2 | 1         |
| <b>Non-Credit Pass/ Fail Type Core Courses</b>                                                                                    |             |                                                        |   |   | <b>0</b>  |
| 1                                                                                                                                 | CHE1001     | Environmental Studies                                  | 2 | 0 | 0         |

|   |         |                                  |   |   |   |
|---|---------|----------------------------------|---|---|---|
| 2 | CEA1001 | Co-/ Extra-curricular Activities | 0 | 0 | 0 |
|---|---------|----------------------------------|---|---|---|

**Program Core (All Courses in this basket are mandatory) - Minimum Credits to be earned from this basket = 41**

|    |         |                                                              |   |   |   |
|----|---------|--------------------------------------------------------------|---|---|---|
| 1  | CSE1006 | Software Engineering                                         | 3 | 0 | 3 |
| 2  | CSE2002 | Web Technology                                               | 2 | 2 | 3 |
| 3  | CSE2045 | Algorithmic Design and Analysis                              | 3 | 2 | 4 |
| 4  | ECE2006 | Computer Architecture and Organization                       | 3 | 0 | 3 |
| 5  | CSE2046 | Operating Systems with Linux Internals                       | 2 | 2 | 3 |
| 6  | CSE2006 | Communication Networks                                       | 2 | 2 | 3 |
| 7  | CSE2007 | Relational Database Management System                        | 2 | 2 | 3 |
| 8  | CSE2008 | Cloud Computing                                              | 2 | 2 | 3 |
| 9  | CSE2009 | Data Analytics and Visualization                             | 2 | 2 | 3 |
| 10 | CSE1035 | Fundamentals of Artificial Intelligence and Machine Learning | 2 | 2 | 3 |
| 11 | MAT2002 | Discrete Mathematics and Graph Theory                        | 3 | 0 | 3 |
| 12 | CSE2015 | Source Code Management                                       | 0 | 2 | 1 |
| 13 | CSE3003 | Cryptography and Network Security                            | 3 | 0 | 3 |
| 14 | CSE3011 | Machine Learning Techniques                                  | 2 | 2 | 3 |

**Discipline Electives - Minimum Credits to be earned from both Common and Specialized elective baskets = 43**

**Common Electives Basket - Minimum Credits to be earned from this basket = 10**

|    |         |                                                |   |   |   |
|----|---------|------------------------------------------------|---|---|---|
| 1  | CSE1007 | Programming in C# and .NET Framework           | 0 | 4 | 2 |
| 2  | CSE1009 | Linux Internals                                | 2 | 2 | 3 |
| 3  | CSE2044 | Computer Graphics with OpenGL                  | 2 | 2 | 3 |
| 4  | CSE2011 | Advanced Java                                  | 0 | 4 | 2 |
| 5  | CSE2012 | Advanced Database Management System            | 2 | 2 | 3 |
| 6  | CSE2013 | Mobile Adhoc Networks                          | 3 | 0 | 3 |
| 7  | CSE2014 | Compiler Design                                | 2 | 2 | 3 |
| 8  | CSE2016 | Software Quality Assurance                     | 2 | 2 | 3 |
| 9  | CSE2017 | UI/UX                                          | 2 | 2 | 3 |
| 10 | CSE2018 | Service-oriented Architecture                  | 3 | 0 | 3 |
| 11 | CSE2019 | Wearable Computing                             | 2 | 2 | 3 |
| 12 | CSE2020 | IT Infrastructure Management                   | 3 | 0 | 3 |
| 13 | CSE2021 | Data Mining and Data Warehousing               | 3 | 0 | 3 |
| 14 | CSE3005 | Wireless Sensor Network                        | 3 | 0 | 3 |
| 15 | CSE3006 | Quantum Computing                              | 2 | 0 | 2 |
| 16 | CSE3007 | AR/ VR/ XR                                     | 2 | 2 | 3 |
| 17 | CSE3008 | Human computer Interaction                     | 3 | 0 | 3 |
| 18 | CSE3009 | Digital Imaging                                | 2 | 2 | 3 |
| 19 | CSE2010 | Mobile Application Development                 | 2 | 2 | 3 |
| 20 | CSE3004 | Parallel and Distributed Computing             | 2 | 2 | 3 |
| 21 | CSE3001 | Theory of Computation                          | 3 | 0 | 3 |
| 22 | CSE3016 | Object Oriented Modelling and Design using UML | 3 | 0 | 3 |
| 23 | MAT2001 | Advanced Statistics                            | 2 | 2 | 3 |

**Specialized Electives Baskets - Minimum Credits to be earned from these baskets (H+I+J+K+L+M+N+O+P)= 33**

**I. AI & ML Basket - Minimum Credits to be earned from this basket = 12**

|   |         |                                             |   |   |   |
|---|---------|---------------------------------------------|---|---|---|
| 1 | CSE3010 | AI & ML Applications                        | 2 | 2 | 3 |
| 2 | CSE3015 | Natural Language Processing                 | 2 | 2 | 3 |
| 3 | CSE3013 | Deep Neural Networks                        | 2 | 2 | 3 |
| 4 | CSE3012 | Optimization Techniques in Machine Learning | 3 | 0 | 3 |
| 5 | CSE3014 | Reinforcement Learning Techniques           | 2 | 2 | 3 |
| 6 | CSE4005 | Industrial Applications of AI & ML          | 3 | 0 | 3 |

**J. Cyber Security Basket - Minimum Credits to be earned from this basket = 3**

|   |         |                                              |   |   |   |
|---|---------|----------------------------------------------|---|---|---|
| 1 | CSE3022 | Intrusion Detection and Prevention System    | 2 | 2 | 3 |
| 2 | CSE3021 | Ethical Hacking                              | 2 | 2 | 3 |
| 3 | CSE3025 | Networking and System Administration         | 2 | 2 | 3 |
| 4 | CSE3020 | Social Media Privacy and Security            | 3 | 0 | 3 |
| 5 | CSE3023 | Fundamentals of Cloud Security               | 3 | 0 | 3 |
| 6 | CSE3024 | Penetration Testing and Incident Response    | 2 | 2 | 3 |
| 7 | CSE3026 | Blockchain and Distributed Ledger Technology | 3 | 0 | 3 |
| 8 | CSE4009 | Industrial Applications of Cyber Security    | 3 | 0 | 3 |

**K. Data Analytics Basket - Minimum Credits to be earned from this basket = 3**

|    |         |                                           |   |   |   |
|----|---------|-------------------------------------------|---|---|---|
| 1  | CSE1008 | Statistical Foundations of Data Analytics | 2 | 2 | 3 |
| 2  | CSE1012 | R Programming for Data Analytics          | 2 | 2 | 3 |
| 3  | CSE1013 | Exploratory Data Analysis                 | 2 | 2 | 3 |
| 4  | CSE2024 | Web Data Analytics                        | 2 | 2 | 3 |
| 5  | CSE2025 | Predictive Analytics                      | 2 | 2 | 3 |
| 6  | CSE2026 | Social Media Analytics                    | 2 | 2 | 3 |
| 7  | CSE2027 | Healthcare Analytics                      | 2 | 2 | 3 |
| 8  | CSE2028 | Big Data Analytics                        | 2 | 2 | 3 |
| 9  | CSE2029 | Business Analytics Fundamentals           | 3 | 0 | 3 |
| 10 | CSE4010 | Industrial Applications of Data Analytics | 3 | 0 | 3 |

**L. IoT Basket - Minimum Credits to be earned from this basket = 3**

|   |         |                                                        |   |   |   |
|---|---------|--------------------------------------------------------|---|---|---|
| 1 | ECE2004 | Sensor Technology, Embedded Systems and User Interface | 2 | 2 | 3 |
| 2 | ECE2002 | IoT Platforms and Application Development              | 2 | 2 | 3 |
| 3 | CSE2031 | Wireless Communication in IoT                          | 3 | 0 | 3 |
| 4 | ECE2001 | IoT Architecture and Protocols                         | 3 | 0 | 3 |
| 5 | CSE2032 | Mobile Application for IoT                             | 2 | 2 | 3 |
| 6 | CSE2033 | Cloud Computing for IoT                                | 3 | 0 | 3 |
| 7 | CSE2030 | Big Data Analytics for IoT                             | 2 | 2 | 3 |
| 8 | ECE2003 | Industrial Internet of Things                          | 3 | 0 | 3 |
| 9 | ECE2005 | Internet of Medical Things                             | 3 | 0 | 3 |

**M. Cloud Computing Basket - Minimum Credits to be earned from this basket = 3**

|   |         |                                            |   |   |   |
|---|---------|--------------------------------------------|---|---|---|
| 1 | CSE2037 | Database and Network Management            | 2 | 2 | 3 |
| 2 | CSE2036 | Cloud Services and APIs                    | 2 | 2 | 3 |
| 3 | CSE2034 | Design and Operation of Data Center        | 3 | 0 | 3 |
| 4 | CSE2076 | System Provisioning and Monitoring         | 2 | 2 | 3 |
| 5 | CSE2038 | Cloud Computing Platforms                  | 2 | 2 | 3 |
| 6 | CSE3028 | Edge Computing Paradigms                   | 3 | 0 | 3 |
| 7 | CSE2039 | Middleware Technologies                    | 3 | 0 | 3 |
| 8 | CSE4003 | Industrial Applications of Cloud Computing | 3 | 0 | 3 |

**N. Robotics Basket - Minimum Credits to be earned from this basket = 3**

|    |         |                                     |   |   |   |
|----|---------|-------------------------------------|---|---|---|
| 1  | ECE5002 | Principles of Robotics and RoS      | 2 | 2 | 3 |
| 2  | ECE2037 | Drone Technology                    | 2 | 2 | 3 |
| 3  | CSE5096 | Robotic Process Automation          | 2 | 2 | 3 |
| 4  | CSE5099 | Robot Motion Planning               | 2 | 2 | 3 |
| 5  | ECE5005 | Robot Kinematics                    | 2 | 2 | 3 |
| 6  | ECE3023 | Mechatronic System Design           | 2 | 2 | 3 |
| 7  | CSE5098 | Computer Vision                     | 2 | 2 | 3 |
| 8  | ECE3026 | Building Robots                     | 0 | 6 | 3 |
| 9  | CSE3031 | Autonomous Mobile Robots            | 2 | 2 | 3 |
| 10 | CSE3032 | Human-robot Interaction             | 2 | 2 | 3 |
| 11 | CSE3033 | Microrobotics                       | 3 | 0 | 3 |
| 12 | CSE4004 | Industrial Applications of Robotics | 3 | 0 | 3 |

**O. Big Data Basket - Minimum Credits to be earned from this basket = 0**

|   |         |                                    |   |   |   |
|---|---------|------------------------------------|---|---|---|
| 1 | CSE3019 | Big Data Security and Privacy      | 3 | 0 | 3 |
| 2 | CSE1011 | Information Visualization          | 2 | 2 | 3 |
| 3 | CSE2080 | Data Engineering Pipelines         | 2 | 2 | 3 |
| 4 | CSE3034 | Data Ingestion Techniques          | 2 | 2 | 3 |
| 5 | CSE4011 | Industrial Applications of BigData | 3 | 0 | 3 |

**P. Full Stack Basket - Minimum Credits to be earned from this basket = 3**

|   |         |                                                     |   |   |   |
|---|---------|-----------------------------------------------------|---|---|---|
| 1 | CSE2022 | Front End Engineering                               | 2 | 2 | 3 |
| 2 | CSE2023 | Backend Engineering                                 | 2 | 2 | 3 |
| 3 | CSE2041 | NoSQL Data Bases and Retrieval Augmented Generation | 2 | 2 | 3 |
| 4 | CSE3017 | Java, Springboot and Microservices                  | 2 | 2 | 3 |
| 5 | CSE3018 | MEAN/ MERN Stack                                    | 2 | 2 | 3 |
| 6 | CSE4008 | Industrial Applications of Full Stack Development   | 3 | 0 | 3 |
| 7 | CSE4012 | Cross Platform Mobile App Development               | 2 | 2 | 3 |
| 8 | CSE4013 | Full Stack AI                                       | 2 | 2 | 3 |

**Q. DevOps Basket - Minimum Credits to be earned from this basket = 3**

|   |         |                                                  |   |   |   |
|---|---------|--------------------------------------------------|---|---|---|
| 1 | CSE2035 | DevOps                                           | 3 | 0 | 3 |
| 2 | CSE2042 | Development Automation                           | 2 | 2 | 3 |
| 3 | CSE2043 | Advanced Linux and Shell Scripting               | 2 | 2 | 3 |
| 4 | CSE2040 | Continuous Integration and Continuous Deployment | 2 | 2 | 3 |
| 5 | CSE2048 | Microservices and Service Oriented Architecture  | 2 | 2 | 3 |
| 6 | CSE2077 | Infrastructure Provisioning                      | 2 | 2 | 3 |
| 7 | CSE2078 | Build and Release Management                     | 2 | 2 | 3 |
| 8 | CSE2079 | Containerization and Orchestration               | 2 | 2 | 3 |
| 9 | CSE4007 | Industrial Applications of DevOps                | 3 | 0 | 3 |

**Open Elective basket - Minimum Credits to be earned from this basket =**

**18**

Any course from **another** 'eligible' curriculum

Any **unused** Discipline Elective from his/ her 'own' curriculum

Any **stand-alone** course declared as "Open Elective" by School/ Dept. for a specific 'eligible' degree program

School approved NPTEL/ Swayam courses for a maximum of 6-credits